

Feasibility Validation Report: Precision Fertilizer Optimization Platform

Executive Summary

Feasibility Assessment: Feasible (Conditional)

Key Finding: The precision fertilizer optimization platform is technically and operationally buildable. MVP (soil data intake → MRTN prescription → VRT export) can launch in 18–24 weeks with a 2–3 person engineering team. Realistic Year 1 target: 10–20 paying customers. Success hinges on three conditions: (1) locking MRTN licensing early, (2) completing VRT export to one OEM brand by Month 4, (3) delivering outcome tracking by Month 6–8 for retention.

MVP Timeline Estimate: 18–24 weeks to production launch with 5–10 beta farmers. Full scale to 10–20 customers by end of Year 1.

Confidence Level: Medium-High (70%) on MVP feasibility. Medium (60%) on Year 1 customer targets (achievable if agronomist partnerships mature and outcome tracking launches on time).

Technical Feasibility

Core Technology Stack: - Backend: Python (Flask/FastAPI) + PostgreSQL + AWS - Frontend: React/Vue for farmer UX - MRTN Model: License Iowa State MRTN (not build) - VRT Export: Native integration to primary OEM (Deere/CNH/AGCO, TBD based on early adopter signal) - Cloud: AWS (ECS + Lambda for variable compute)

Technical Complexity: Medium

Critical Path: - Weeks 1–4: OEM API learning curve + VRT integration design - Weeks 4–10: VRT export build + MRTN wrapper implementation - Weeks 10–16: Integration testing + edge case handling - Weeks 16–20: Beta testing with 2–3 cooperative farmers - Weeks 20–24: Production hardening + launch

Technical Risks & Mitigations: - VRT API complexity (HIGH): Each OEM has different formats/authentication. Mitigation: Start with one brand. Test on actual farmer equipment (5+ farms) before launch. Assign one engineer to own this. - Soil data normalization (MEDIUM): Farmers have messy soil test data from multiple labs. Mitigation: Manual intake form (not auto-parsing). Validate data quality before MRTN processing. - MRTN edge cases (MEDIUM): Model assumes standard corn-on-corn rotation. Mitigation: Build guardrails to flag non-standard scenarios. Document limitations.

Scalability Assessment: MVP architecture scales to 100+ farms without redesign. Cloud infrastructure (AWS Lambda, PostgreSQL) grows with volume. Biggest bottleneck: VRT export reliability and OEM API maintenance.

Operational Feasibility

Key Operations: - Soil data intake: Farmers upload test results via form (manual entry or file upload). Team validates/normalizes (2–4 hours per farm). - Prescription generation: Automated MRTN calculation + commodity price integration. Real-time or batch processing. - VRT export: System generates OEM-specific file format. Farmer downloads and imports to equipment. - Agronomist review: Year 1 includes optional agronomist co-sign (liability buffer, builds trust).

Resource Requirements: - Team: 2–3 engineers (backend, frontend, infrastructure) + 1 product manager + 1 customer success (hire Month 3–4) - Tools: Airtable/Typeform (data intake), AWS (infrastructure), OEM APIs (VRT export) - Year 1 staffing budget: \$400–550K (team + operations)

Bottlenecks Identified: - Customer onboarding (40–80 hours Year 1): Soil data cleanup is manual and time-intensive. Solution: Build validation workflows. Flag bad data early. Reject submissions below quality threshold. - Agronomist adoption (slow): Agronomists won't endorse without proof. Solution: Frame as co-development. Offer 6-month co-pilot period. Document agronomist feedback. - Outcome tracking dependency (CRITICAL): Without ROI proof by Month 6–8, Year 2 retention and expansion suffer. Solution: Lock outcome tracking timeline now. Build yield data integration architecture in parallel (don't wait until Month 6).

Build vs Buy Analysis

Build (Your IP Layer): - Prescription workflow: Soil → MRTN + pricing logic → farmer-facing recommendations - ROI tracking dashboard: Field outcome measurement, cost/yield attribution - Farmer UX: Intuitive onboarding, prescription visualization - Agronomist partnership platform: Co-development, co-sign, communication tools

Buy/Integrate (Third-Party): - MRTN model: License Iowa State (NOT build proprietary model Year 1). Cost: \$5–10K/year. Timeline: Lock by Month 1. - Soil data: Manual intake form + integration with Ward Labs/Soli/A&L APIs (Phase 2, not MVP). - Weather/commodity feeds: USDA NASS (free), Weather Underground (\$500–2K/year). - VRT export: Native OEM API (Deere, CNH, AGCO). Start with one. Expand based on customer demand. - Cloud: AWS (infrastructure as a service).

Partner (Co-Development/Distribution): - University (Iowa State/Illinois): MRTN licensing + field trial validation (co-publish results) - Agronomists: White-label platform, co-validation, reference accounts (not reseller model) - Soil test labs: Data sharing partnerships (Year 2+, not MVP)

Integration Complexity: Medium Multiple APIs but low interdependency risk. Each integration can

progress independently. OEM VRT API is critical path (tightly couples to customer launch).

Year 1 API/Integration Budget: - MRTN licensing: \$5–10K - Weather + commodity data: \$1–2K - Cloud infrastructure: \$30–60K - OEM API integrations (engineering): \$20–35K (embedded in team) - Contingency (unexpected partnerships): \$5–10K - **Total: \$60–120K**

Compliance & Regulatory

Regulatory Requirements: - Nutrient management: MRTN model is university-validated and compliant with Iowa, Illinois, Indiana state guidelines. Audit model against state-specific regulations (NRCS, WOTUS rule) by Month 1. - Data privacy: CCPA and GDPR considerations. Farmer data is proprietary. Build data encryption + ownership clauses into Terms of Service. - Liability: Recommendations are advisory. Farmer and agronomist own final decision. Document agronomist co-sign in writing. Carry E&O insurance (\$3–5K/year).

Data Privacy: - Encrypt field data at rest (AWS KMS) and in transit (TLS 1.3) - Farm data ownership: Farmer owns all field data; can export/delete at any time - No third-party data sharing without explicit consent - Build audit trail for all recommendations and co-signs

Compliance Path: - Month 1: Draft Terms of Service + Privacy Policy (\$3–5K legal review) - Month 1: Audit MRTN against state regulations (\$2–3K agronomist review) - Month 2: Secure E&O insurance (\$3–5K/year) - Month 2: Document data security practices (\$2–3K) - Ongoing: Monitor EPA/state nutrient regulations (budget \$1–2K/year monitoring)

Year 1 Compliance Budget: \$12–16K

Expert Debate Insights

Areas of Consensus: - Problem is real and painful (fertilizer cost volatility, over-application suspicion validated by customer interviews) - One OEM brand at MVP launch reduces risk significantly (vs. attempting all three simultaneously) - Manual soil data intake is correct call (vs. building fragile PDF parsers) - Deferring outcome tracking to Month 6–8 is pragmatic (don't over-engineer MVP) - Agronomist co-sign in Year 1 is liability buffer and credibility signal (essential for early trust)

Areas of Debate Resolution: - VRT Integration Timeline: Experts initially estimated 8 weeks; refined to 16–20 weeks with production hardening and real equipment testing. Consensus: 5-week learning curve per OEM brand is realistic. - Year 1 Customer Target: Viability Report projected 20–30 customers. Realistic assessment: 10–20 customers (aggressive to hit 20–30 if agronomist channel underperforms). Consensus: Target 10–20, stretch to 25–30 if traction strong. - Outcome Tracking Launch: Month 6 is ideal, Month 8 acceptable. Slip beyond Month 8 risks Year 2 retention. Consensus: Lock timeline by Month 2.

Unresolved Questions: - Which OEM brand first? (Founder resolving via early adopter signal, not arbitrary choice) - Will agronomists actively partner or passively observe? (Will become clear Month 3–4) - Can outcome tracking achieve statistical rigor by Month 6–8? (Depends on data quality + engineering capacity)

Implementation Roadmap

Phase 1 – Foundation & MVP Build (Weeks 1–24): - Weeks 1–2: Secure MRTN licensing, get OEM API access, finalize legal docs - Weeks 2–4: Build soil intake form, define MRTN wrapper architecture - Weeks 4–10: VRT export integration, MRTN + pricing logic, farmer UX - Weeks 10–16: Integration testing, edge case handling, production hardening - Weeks 16–20: Beta testing (2–3 farms), fix bugs, validate equipment compatibility - Weeks 20–24: Production launch, 5–10 early farmers live

Phase 2 – Growth & Outcome Tracking (Months 6–12): - Month 6: Outcome tracking MVP (yield data integration, cost/savings calculation) - Months 6–8: Agronomist co-development partnerships (3–5 active pilots) - Months 6–12: Scale to 10–20 customers, build case studies - Month 9+: Begin Year 2 proprietary model planning (if outcome data supports)

Critical Path Items (Must Execute on Time): - MRTN licensing (Month 1): Delays block MRTN integration and cost architecture - OEM VRT API integration (Weeks 4–10): Delays push launch 4–8 weeks - Outcome tracking launch (Month 6–8): Delays risk Year 2 retention and expansion - Agronomist recruitment (Month 2–4): Delays impact Year 2 growth channel

Contingency (If Outcome Tracking Slips): - Month 6–8 slip acceptable if you have 3+ published case studies showing ROI proof from early customers - Slip beyond Month 8 requires messaging pivot: reframe as "transparent methodology" instead of "verified ROI"

Recommendations

Immediate Actions (Weeks 1–4): 1. Contact Iowa State + University of Illinois for MRTN licensing terms/pricing — finalize deal by end of Month 1 2. Get OEM API sandbox access for primary brand (Deere, CNH, or AGCO per early adopter signal) 3. Confirm engineering team composition (2–3 engineers minimum). Identify one engineer to own VRT integration (critical path) 4. Hire/queue legal review for Terms of Service, Privacy Policy, E&O insurance quotes 5. Identify 5–10 pilot farmers with target OEM equipment brand

Before Building (Month 1–2): - Audit MRTN model against state nutrient regulations (Iowa, Illinois, Indiana) - Finalize liability framework: farmer/agronomist ownership, platform role as advisory - Build soil data intake validation rules (what data quality is acceptable?) - Document OEM API requirements + version control strategy (how to handle API changes?) - Recruit 3–5 progressive agronomists as co-developers (not resellers)

During Build (Months 2–4): - Ship soil intake form + MRTN wrapper (testable by Month 3) - Integrate VRT export + begin equipment compatibility testing - Hire customer success person (Month 3–4 start date) - Begin outcome tracking architecture design (yield data sources, cost calculation logic)

Team Priorities: 1. VRT export reliability (customer-facing, trust-critical) 2. Soil data quality management (operational bottleneck) 3. Outcome tracking architecture (retention lever, not MVP but prep now) 4. Agronomist partnership pipeline (distribution channel)

Risk Summary & Path Forward

Highest Risk Factors: - VRT export integration complexity: Each OEM API is proprietary. Test with real equipment required. Mitigation: Start with one brand. Assign dedicated engineer. Test early and often. - Outcome tracking delay: If slips beyond Month 8, Year 2 renewal suffers. Mitigation: Lock timeline Month 2. Build architecture now (don't defer). Have agronomist co-sign as interim retention strategy. - Agronomist adoption slower than projected: Agronomists are risk-averse, slow to endorse. Mitigation: Frame as co-development, not reseller. Expect 12–18 month relationship building. Don't rely on agronomist channel for Year 1 acquisition.

Go/No-Go Decision Points: - End of Month 1: MRTN licensing finalized? OEM API access confirmed? Legal docs drafted? (All must be YES to proceed) - End of Month 4: VRT export to one brand working? (Must be YES for launch on schedule) - End of Month 6: 5–10 early customers live? Outcome tracking roadmap clear? (Must be YES for Year 2 growth)

Overall Feasibility Verdict: FEASIBLE. You have strong team, sound technical approach, and realistic expectations. Success depends on executing the critical path (VRT export, outcome tracking, agronomist partnerships) on time and learning fast from early customers. No technical blockers; primary risk is execution speed and operational discipline.